

Assignment 9

Due Sunday November 16th, 2025, Self Grading Due Sunday November 23rd, 2025

1. Review Madhow Chapter 6.
2. Madhow Problem 6.20a,c
3. A communication link uses PAM3 (three symbols to encode information), and the received signal is modeled as,

$$Y = X + N,$$

where X can take the values of $\{-a, 0, a\}$ and N is Gaussian noise with a variance of N_0 . Each symbol is equally likely.

- (a) State the three hypotheses to test. Draw the decision regions for the three symbols.
 - (b) Compute the probability of error. Express the result in terms of Q .
4. With FSK, different frequencies are used to represent the data bits and the transmitted signal for $0 \leq t \leq T$ is

$$s(t) = \begin{cases} A \cos(2\pi f_1 t), & 0 \leq t \leq T \text{ if a 0 is sent} \\ A \cos(2\pi f_2 t), & 0 \leq t \leq T \text{ if a 1 is sent} \end{cases}.$$

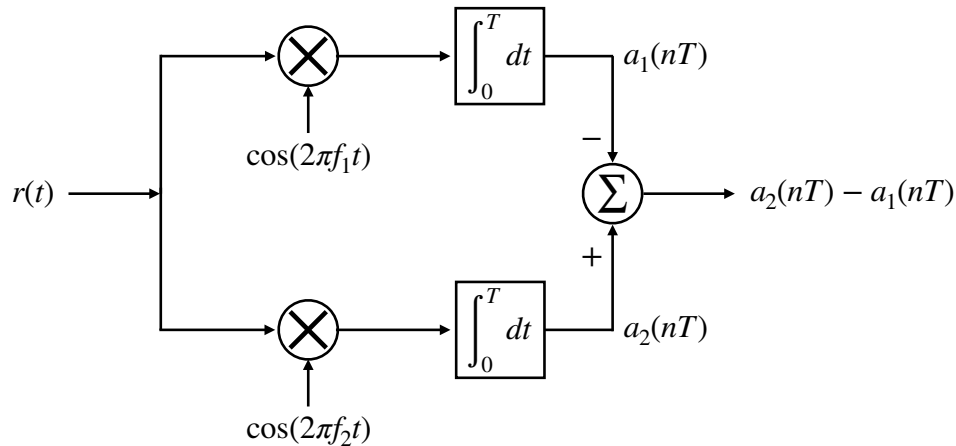
The frequencies f_1 has K cycles in time T and f_2 has $2K$ cycles in time T ensuring that the corresponding waveforms are orthogonal,

$$\int_0^T \cos(2\pi f_1 t) \cos(2\pi f_2 t) dt = 0.$$

The received signal is,

$$r(t) = s(t) + n(t)$$

where $n(t)$ is Gaussian Noise with zero mean and variance N_0 . The received signals are passed through a coherent detector with two correlators as shown below.



- (a) What is the distribution of the noise after the correlator? **Hint:** Look at Example 5.8.2 in Madhow.

$$\int_0^T n(t) \cos(2\pi f_1 t) dt$$

- (b) Suppose a 1 is transmitted, what's the distribution of the matched filter outputs (a_1 and a_2)?
- (c) What's the distribution of $a_2(nT) - a_1(nT)$? Compute the probability of error. **Hint:** The noise from a_1 and a_2 are uncorrelated, use that result to find the variance of $a_2 - a_1$.
- (d) The output of the detector is $a_2 - a_1 = -0.1$. Assuming that $AT/2 = 1$, $N_0T = 1$, and f_2 comes 75% of the time, use a soft decision to determine the bit.

5. **Self-grade Homework 8.** Fill out the self-grade rubric for Homework 8. Attach to Homework 9 submission.